Questions: Introduction to algebraic fractions

Donald Campbell

Summary

A selection of questions for the study guide on the introduction to algebraic fractions.

Before attempting these questions, it is highly recommended that you read Guide: Introduction to algebraic fractions. You may also find reading Guide: Factorization useful.

Q1

For each algebraic fraction, state the restriction(s) on the denominator.

1.1. $\dfrac{3}{x}$

1.2. $\dfrac{5}{x-4}$

1.3. $-\dfrac{2}{2x+1}$

1.4. $\dfrac{y+3}{y+5}$

1.5. $\dfrac{4x}{x(3x-2)}$

1.6. $\dfrac{t-1}{(t+2)(t-3)}$

1.7. $\dfrac{5x+1}{x^2-16}$

1.8. $\dfrac{2}{(x-1)^2}$

1.9. $\dfrac{b}{b^2+4b}$

1.10. $\dfrac{7}{(x+3)(2x-5)}$

1.11. $-\dfrac{x-4}{x^2-5x+6}$

1.12. $\dfrac{z+3}{z^2+2z-8}$

Q2

In each of the following, find the missing value ? that makes the two algebraic fractions equivalent.

2.1. $\dfrac{x}{3} = \dfrac{?}{9}$

2.2. $\dfrac{2}{t} = \dfrac{10}{?}$

2.3. $\dfrac{a+1}{4} = \dfrac{?}{12}$

2.4. $\dfrac{3x}{5} = \dfrac{?}{20}$

2.5. $\dfrac{5}{x+2} = \dfrac{?}{2(x+2)}$

2.6. $\dfrac{t-3}{t} = \dfrac{?}{2t}$

2.7. $\dfrac{4}{y-1} = \dfrac{12}{?}$

2.8. $\dfrac{x}{x-4} = \dfrac{?}{3(x-4)}$

2.9. $\dfrac{x+5}{2x} = \dfrac{3(x+5)}{?}$

2.10. $\dfrac{2z-1}{z+3} = \dfrac{?}{3(z+3)}$

2.11. $\dfrac{3}{x} = \dfrac{3(x+1)}{?}$

2.12. $\dfrac{r}{r-2} = \dfrac{r(r+3)}{?}$

2.13. $\dfrac{4x}{x+1} = \dfrac{?}{(x+1)^2}$

2.14. $\dfrac{2x+1}{x-5} = \dfrac{6x+3}{?}$

2.15. $\dfrac{5-z}{z+2} = \dfrac{?}{-2(z+2)}$

Q3

Simplify each algebraic fraction.

3.1. $\dfrac{8y}{12y}$

3.2. $\dfrac{5x^2}{15x}$

3.3. $\dfrac{n^2+4n}{n}$

3.4. $\dfrac{3x^2-9x}{6x}$

3.5. $-\dfrac{4x}{10x}$

3.6. $\dfrac{m^2-16}{m-4}$

3.7. $\dfrac{x^2-1}{x+1}$

3.8. $\dfrac{x^2+5x+6}{x+2}$

3.9. $\dfrac{z^2+3z-10}{z-2}$

3.10. $\dfrac{2x^2+7x+3}{x+3}$

3.11. $\dfrac{y^2-10y+25}{y-5}$

3.12. $\dfrac{4-x}{x-4}$

3.13. $\dfrac{2x^2-8}{x^2-4}$

3.14. $\dfrac{c^3-4c}{c^2-4}$

3.15. $\dfrac{A^2-3A-10}{A^2-25}$

After attempting the questions above, please click this link to find the answers.

Version history and licensing

v1.0: initial version created 12/25 by Donald Campbell as part of a University of St Andrews VIP project.

This work is licensed under CC BY-NC-SA 4.0.